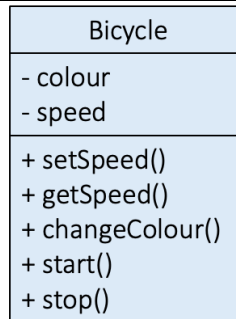


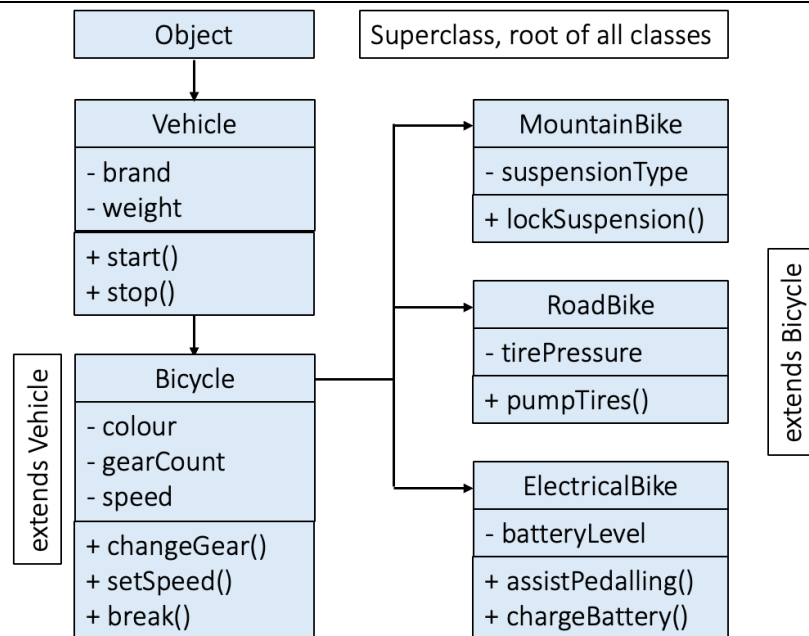
ASSIGNMENTS – OOADI

LECTURE 3

1.



2.



3.

```
class Vehicle {
    // Superclass is Vehicle
    protected String brand;
    protected double weight;

    public void start() {
        System.out.println("Vehicle started");
    }

    public void stop() {
        System.out.println("Vehicle stopped");
    }
}
```

```
class Bicycle extends Vehicle {
    // Sub-class extension of Vehicle and superclass to following types
    protected String colour;
    protected int gearCount;
    protected int speed;

    public void changeGear(int g) {
        gearCount = g;
    }
    public void setSpeed(int s) {
        speed = s;
    }
}

class MountainBike extends Bicycle {
    private String suspensionType = "full";

    public void lockSuspension() {
        System.out.println("Suspension locked");
    }
}

class RoadBike extends Bicycle {
    private double tirePressure = 8.0;

    public void pumpTires() {
        System.out.println("Tires pumped to " + tirePressure + " bar");
    }
}

class ElectricBike extends Bicycle {
    private int batteryLevel = 80;

    public void assistPedalling() {
        System.out.println("Pedal assist on (" + batteryLevel + "%)");
    }
    public void chargeBattery() {
        batteryLevel = 100;
    }
}
```

4. package lecture3;

```
public class Assignment4 {
    public static void main(String[] args) {

        MountainBike mb = new MountainBike();
```

```
mb.start();           // from Vehicle
mb.setSpeed(20);      // from Bicycle
mb.lockSuspension();  // from MountainBike

RoadBike rb = new RoadBike();
rb.start();
rb.pumpTires();       // subclass-specific

ElectricBike eb = new ElectricBike();
eb.start();
eb.assistPedalling(); // subclass-specific
}
}
```

5. The keyword 'this' refers to the current object (the instance itself). It is used for:

- Accessing current object's attributes
- Calling current object's methods
- Distinguishing between parameters and fields (this.colour = colour;)

The 'super' keyword refers to the parent class of the current object and is used for:

- Calling superclass methods (super.start();)
- Calling superclass constructor (super(brand, weight);)
- Accessing hidden fields

And to the question whether it would make sense to have a keyword 'sub', then the answer is no. And this comes on the basis of several reasons:

- The superclass must not depend on knowledge about its subclasses (violates OO principles)
- Subclasses are allowed to extend superclasses freely, and the superclass should not need to know who extends it
- Having a sub keyword would invert inheritance direction and break encapsulation.

Therefore, having a 'sub' keyword is not meaningful in proper object-oriented design.
